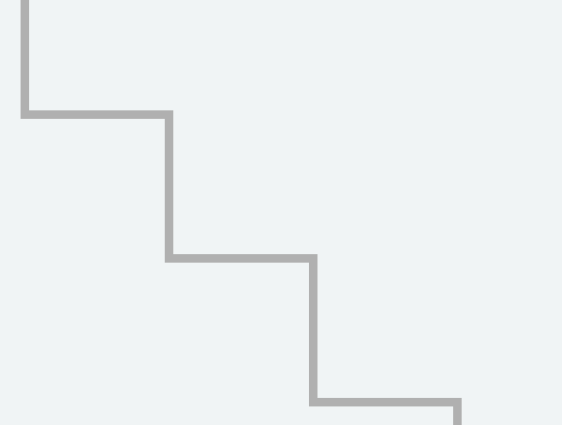
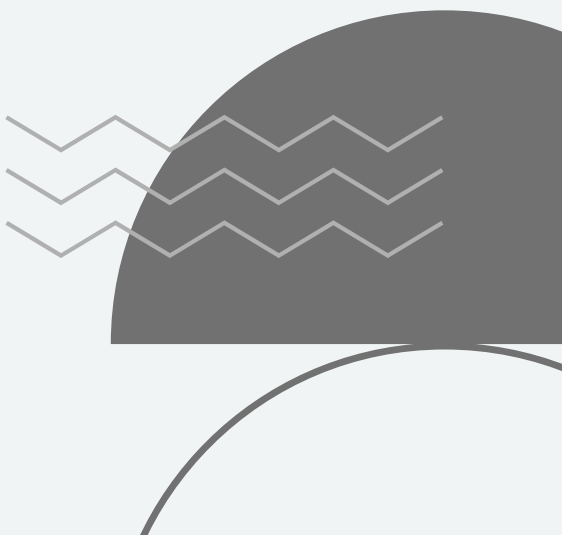
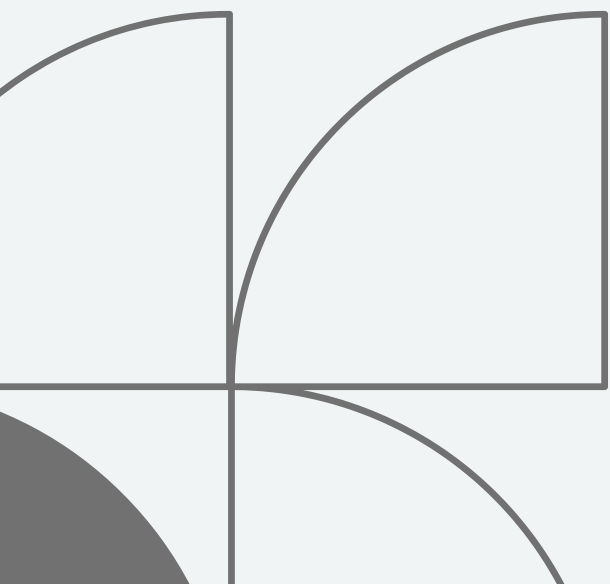




STUDY AND IMPLEMENTATION OF A SWITCHED-CAPACITOR BASED FLOATING INVERTER AMPLIFIER

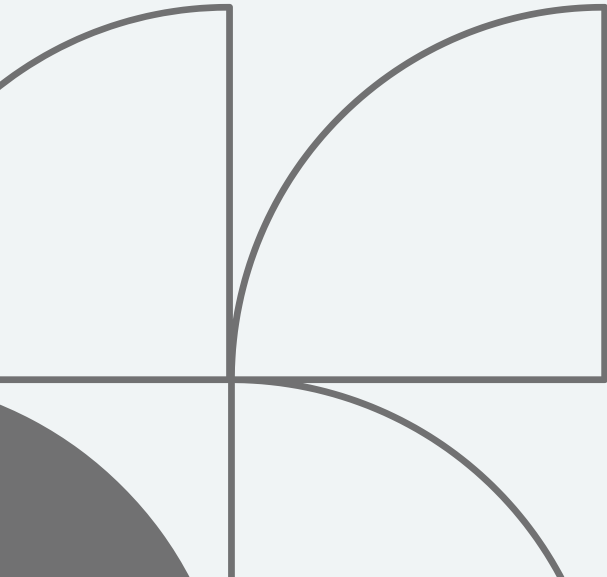
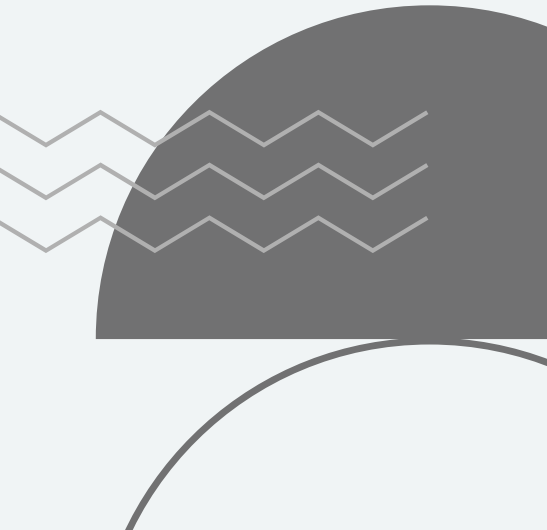
Presented By Ishita Porwal

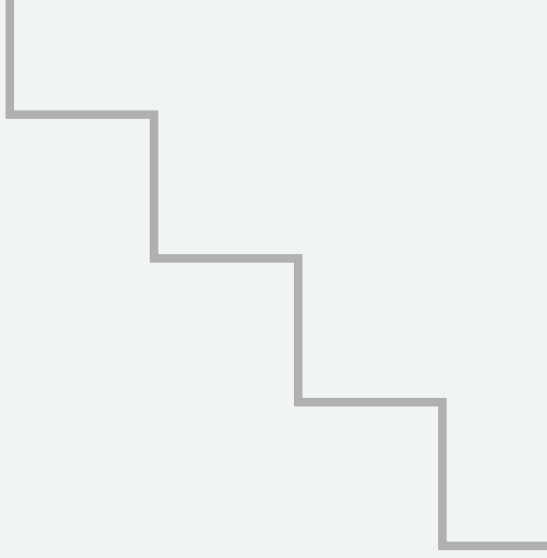
Under Mentorship of Professor R. S. Ashwin Kumar



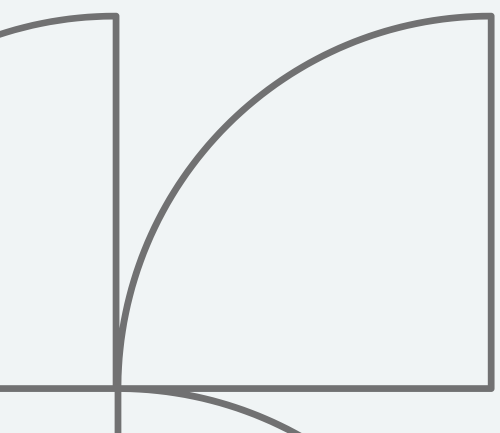
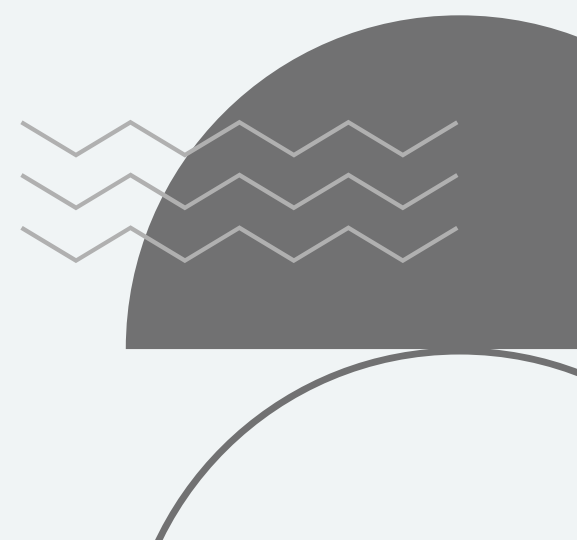


MOTIVATION & PROBLEM STATEMENT

- Continuous CMOS scaling drives supply voltages lower, severely limiting the voltage headroom of traditional operational amplifiers in modern switched-capacitor (SC) networks.
 - The Floating Inverter Amplifier (FIA) emerges as a highly effective, low-voltage alternative.
 - It operates without stacked tail-current sources, achieving high energy efficiency and a compact footprint.
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PROJECT METHODOLOGY

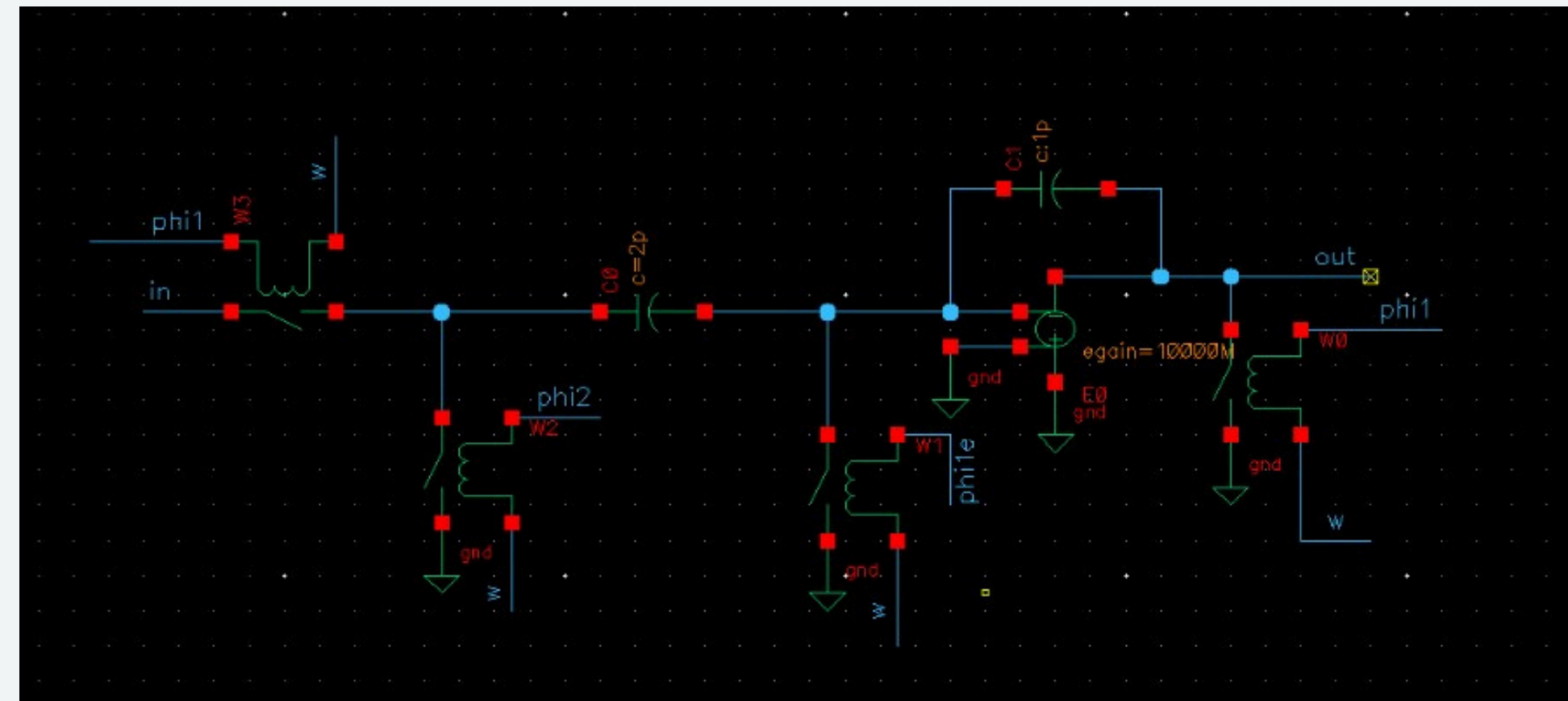
- **Phase 1 – Ideal Baseline:** Design of a switched-capacitor amplifier using ideal switches and an ideal VCVS to establish reference performance
 - **Phase 2 – FIA Characterization:** Detailed analysis of the FIA, including voltage gain and variation of $g_{m,avg}$ with transistor fingers and reservoir capacitance
 - **Phase 3 – System Integration:** Replacement of the ideal VCVS with the FIA in a differential SC topology to realize a practical circuit
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SC CIRCUIT OPERATION

Understanding the foundational principles

Operation

Two-phase operation uses strictly non-overlapping clocks (Φ_1 , Φ_2) to transfer discrete charge packets, enabling accurate sampling and amplification without the need for resistors. During the sampling phase (Φ_1 HIGH), the input signal is stored on the capacitor C_i , while the feedback capacitor C_f is reset to remove any residual charge. An early clock phase (Φ_{1e}) is used for bottom-plate sampling, which reduces charge injection. In the amplification phase (Φ_2 HIGH), the switches reconfigure the circuit so that the stored charge on C_i is transferred to C_f through the virtual ground of the VCVS. This charge redistribution results in an output voltage $V_{out} = -(C_i/C_f) \cdot V_{in}$, demonstrating gain determined purely by the capacitor ratio.



OUTPUT WAVEFORM



Simulated response showing the input (green) and corresponding output wave form (yellow).
The output correctly follows the inverted, scaled input with gain 2



FLOATING INVERTER AMPLIFIER

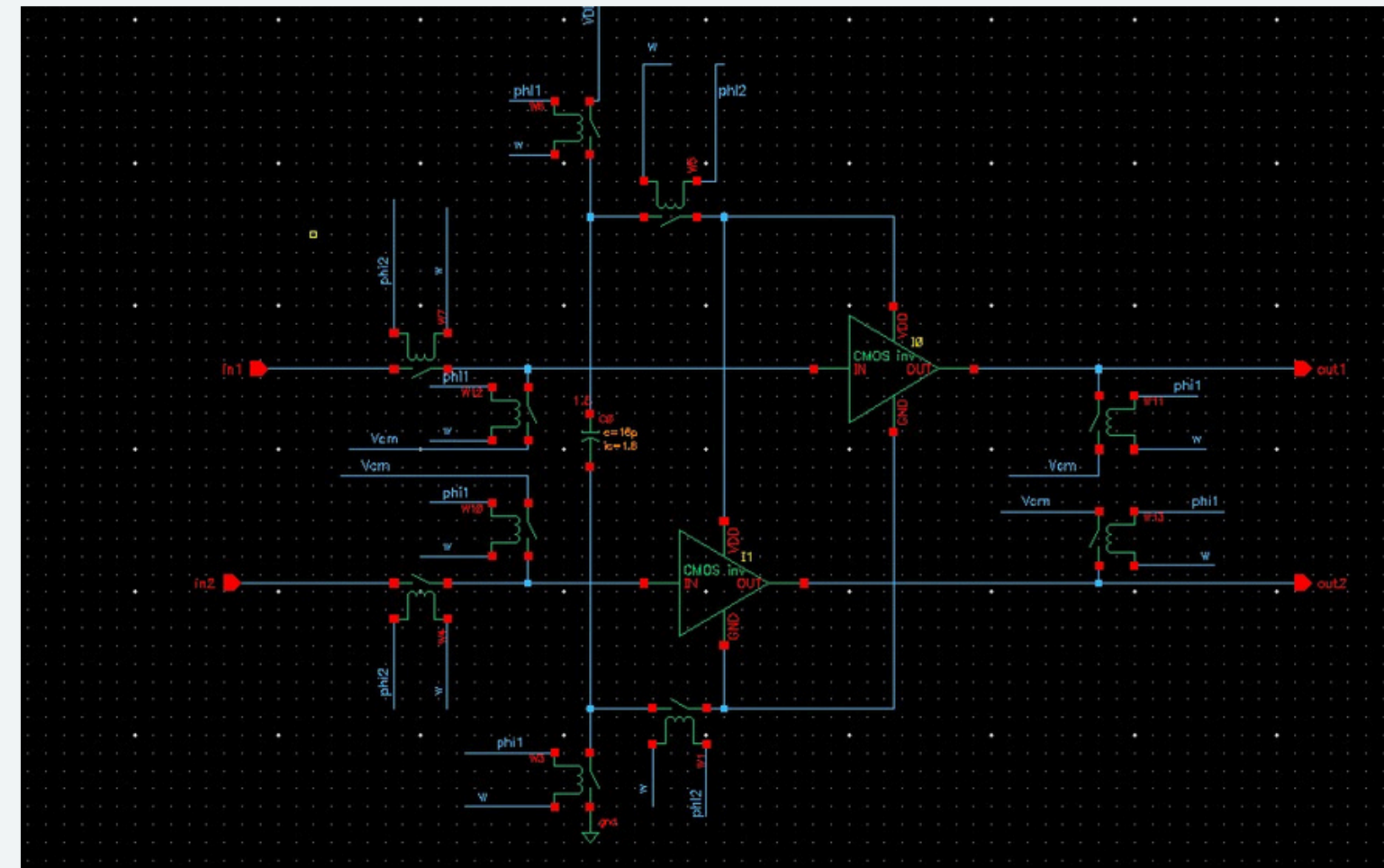
Understanding the FIA

Structure

The Floating Inverter Amplifier features a standard CMOS inverter design with isolated floating supply rails, enhancing performance and efficiency in low-voltage applications.

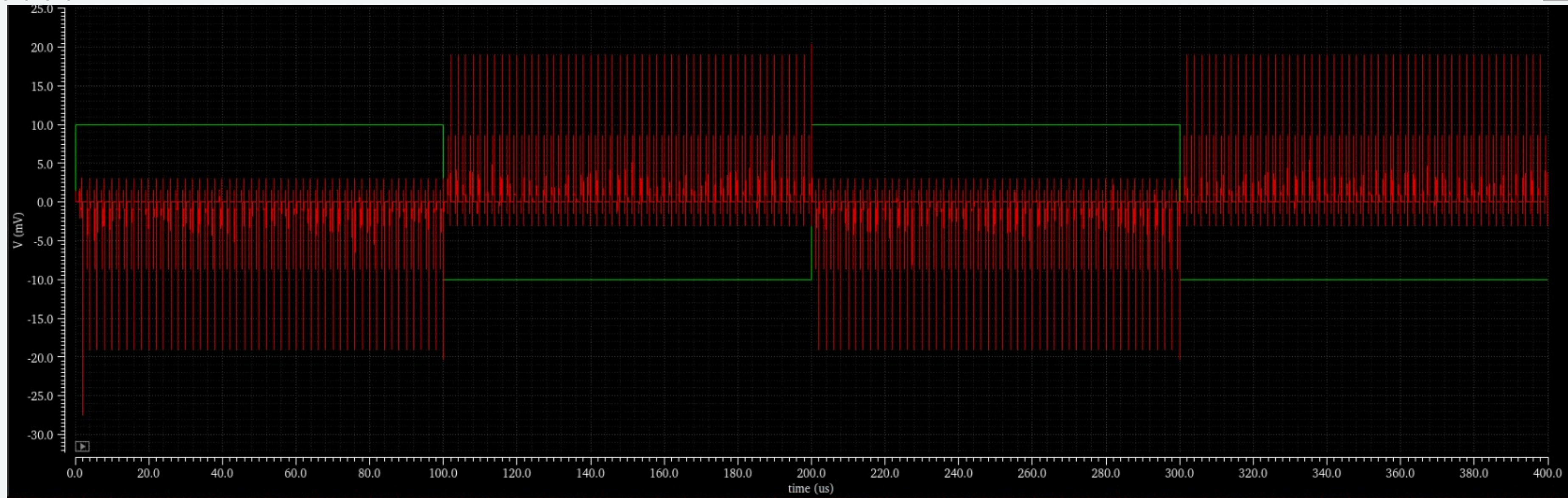
Operation

It operates using a precharged reservoir capacitor, enabling both NMOS and PMOS devices to contribute simultaneously to output current, boosting overall transconductance and gain.



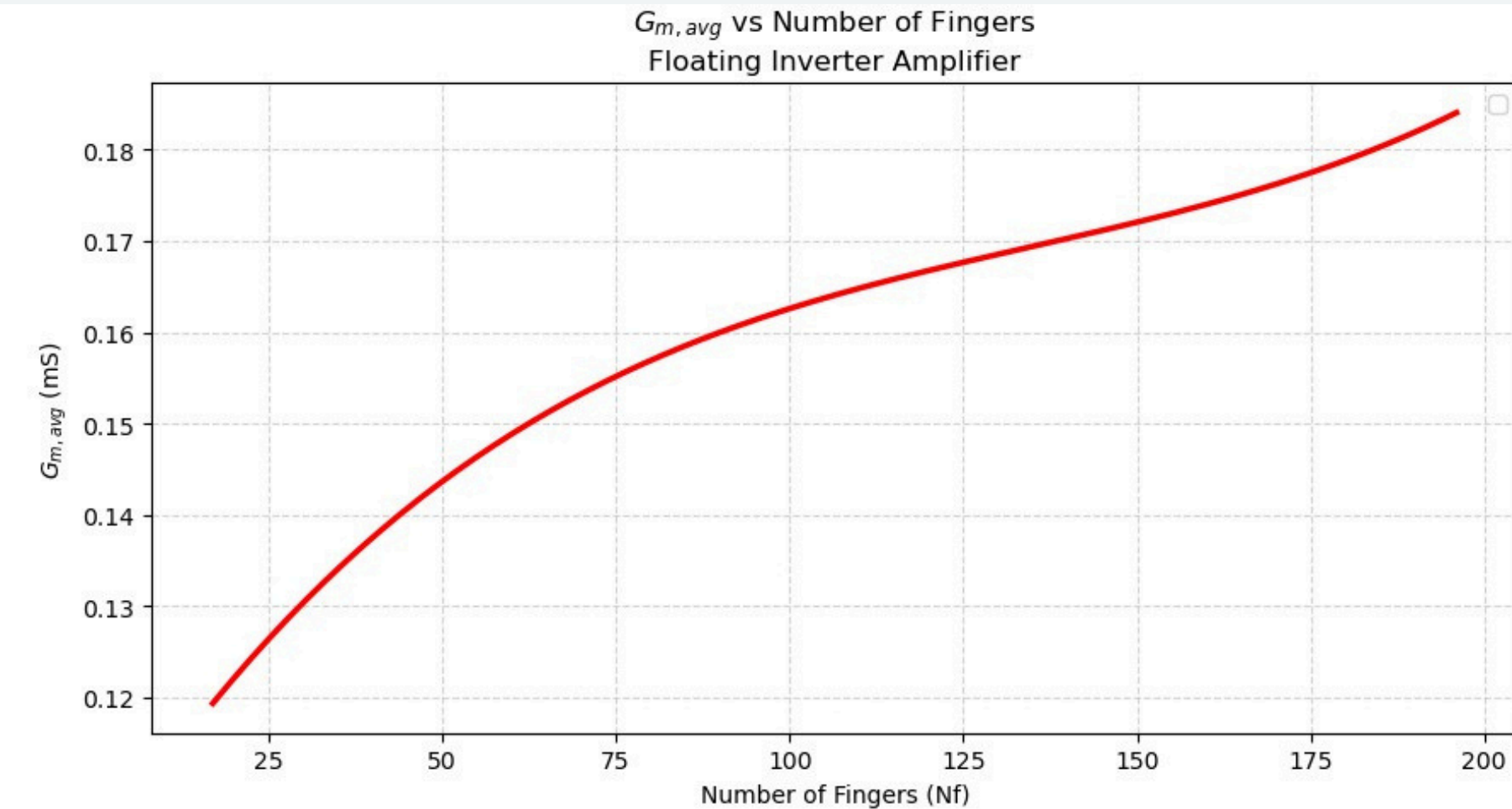
Design from X. Tang et al., “A 13.5b-ENOB second-order noise-shaping SAR with PVT-robust closed-loop dynamic amplifier,” in IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers, Feb. 2020, pp. 162–164.

OUTPUT WAVEFORM

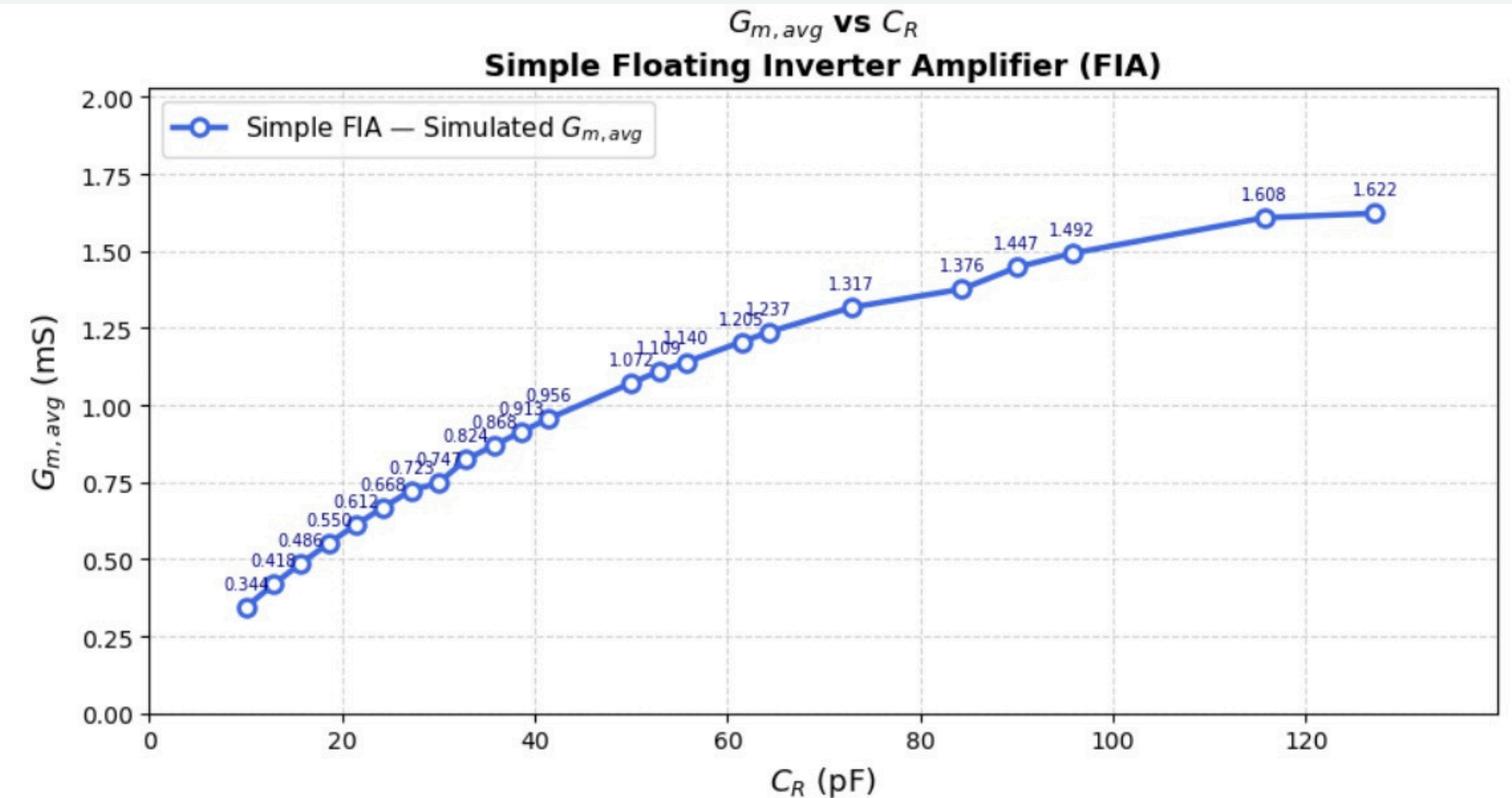


Transient simulation of the fully differential FIA. The green trace represents the 10 mV differential input pulse, and the red trace shows the resulting inverted differential output, settling to approximately 22 mV during the active phase.

VARIATION OF AVERAGE TRANSCONDUCTANCE WITH FINGERS AND RESERVOIR CAPACITANCE



The number of fingers (Nf) of MN and MP was swept (15–200) with constant finger width to study sizing effects. The average transconductance g_m increased nearly linearly up to ~60 fingers, then deviated. Time-varying g_m during Φ_2 was averaged and analyzed



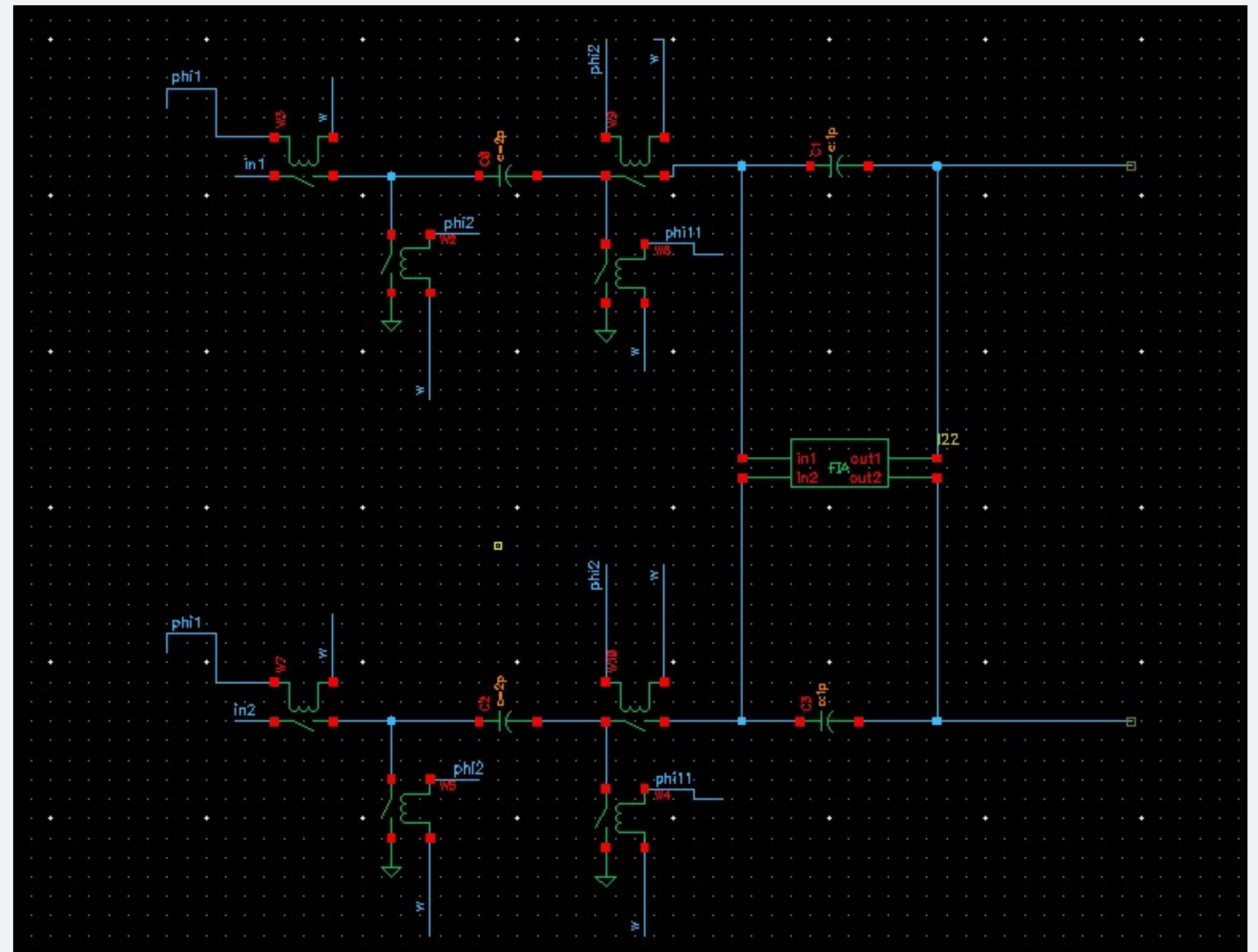
A parametric sweep of the reservoir capacitor C_R shows that increasing C_R reduces voltage droop and increases $g_{m,avg}$, stabilizes the inverter operating point, and results in higher average transconductance $g_{m,avg}$ during Φ_2 .

Results were compared with results in Analysis and Design of a Discrete-Time Delta-Sigma Modulator Using a Cascoded Floating-Inverter-Based Dynamic Amplifier

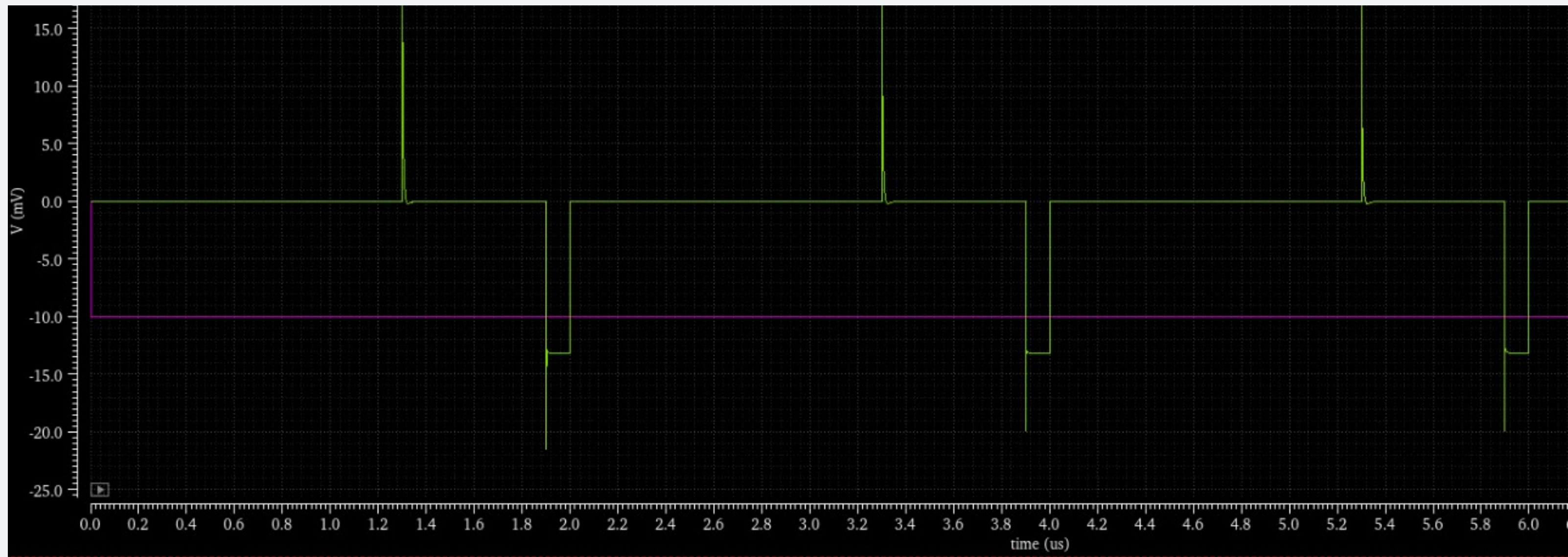
DIFFERENTIAL SC AMPLIFIER WITH FIA

Operation

The circuit operates in two phases. During the sampling phase ($\Phi 1$ HIGH), the differential input is stored on input capacitors, with $\Phi 1 1$ turning LOW slightly earlier to trap accurate charge and reduce injection errors. In the amplification phase ($\Phi 2$ HIGH), the input nodes are grounded, transferring the stored charge to the feedback capacitors. The FIA, supported by its reservoir capacitor, maintains virtual ground and generates a 2 V/V output.



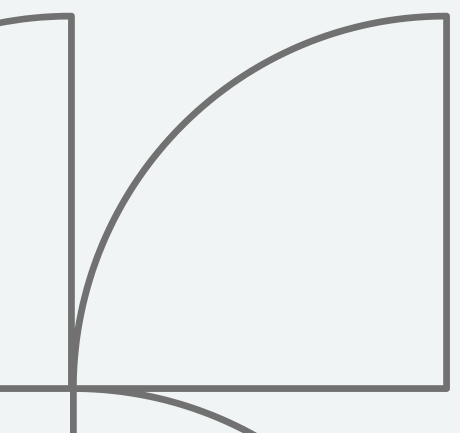
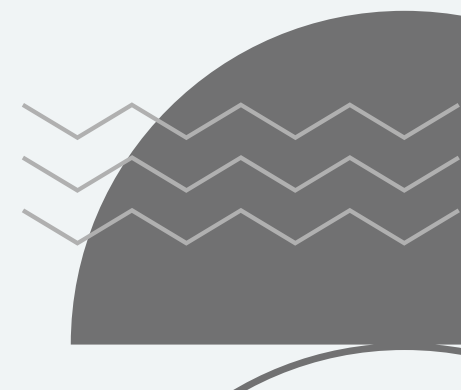
OUTPUT WAVEFORM



A negative input produces a negative output, confirming an overall non-inverting behavior due to two inherent inversions (charge transfer and inverter action). The simulated gain ($\sim +1.3$ V/V) is reduced due to finite FIA gain and parasitic capacitances. Despite this, the circuit achieves stable, power-efficient discrete-time amplification.

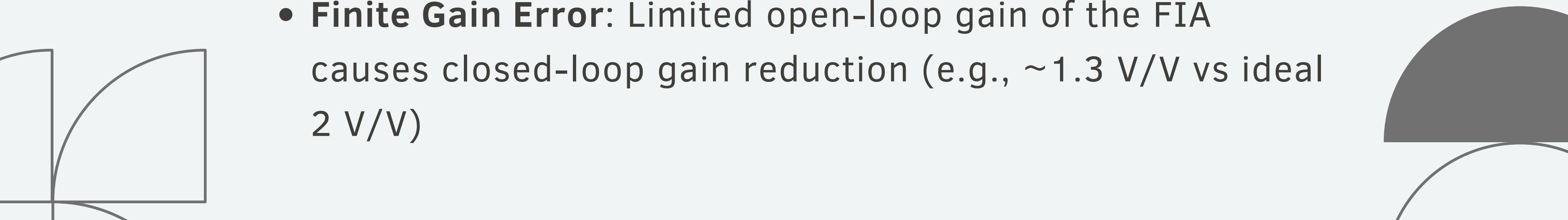


ADVANTAGES OF SC-FIA OVER CONVENTIONAL SC-OTA

- **Higher Transconductance Efficiency:** Effective $g_m = g_{m,n} + g_{m,p}$ giving higher transconductance per unit current than a single differential pair
 - **Low-Voltage Operation:** No tail current source → reduced stacking, improved output swing, and better operation at low V_{DD}
 - **Improved Power Efficiency:** Floating reservoir capacitor (CR) enables partial energy recycling, enhancing dynamic power efficiency
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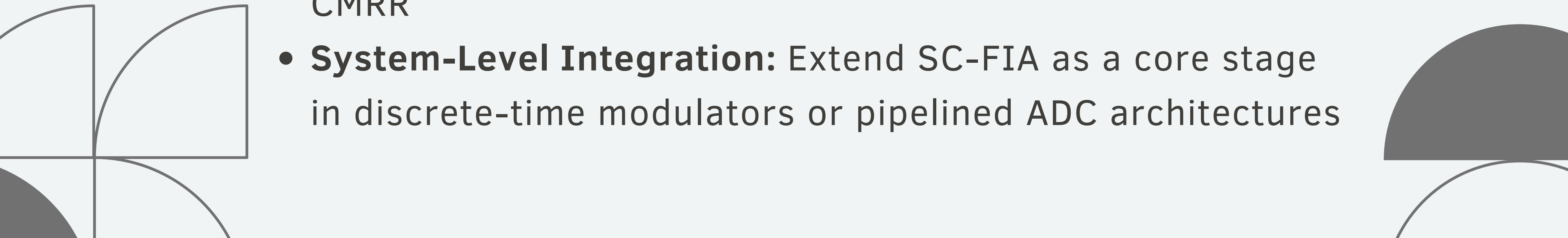


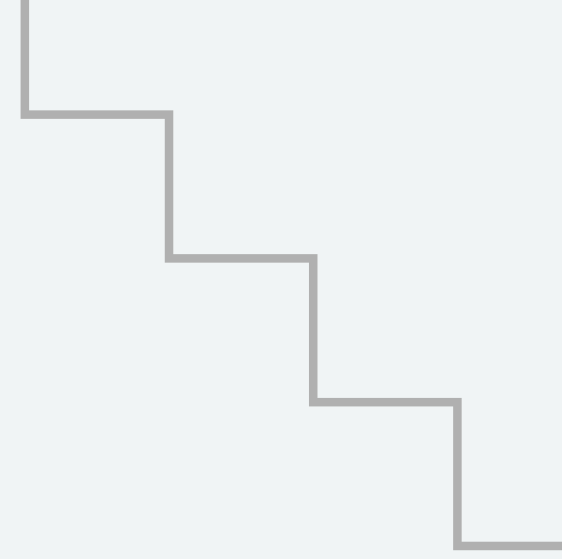
OBSERVED NON-IDEALITIES

- **Sub-linear gm Scaling:** $g_{m,avg}$ increases sub-linearly at high finger count due to rising parasitic gate resistance and C_{gs} , limiting bandwidth
 - **Reservoir Cap Saturation:** Improvement in $g_{m,avg}$ saturates beyond ~ 100 pF; further increase adds area with negligible benefit
 - **Finite Gain Error:** Limited open-loop gain of the FIA causes closed-loop gain reduction (e.g., ~ 1.3 V/V vs ideal 2 V/V)
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FUTURE IMPROVEMENTS

- **Gain Boosting:** Apply cascoding (if voltage headroom allows) or active gain-boosting to enhance open-loop gain and reduce closed-loop error
 - **Offset Compensation:** Incorporate digital calibration or discrete-time trimming to correct offsets and improve CMRR
 - **System-Level Integration:** Extend SC-FIA as a core stage in discrete-time modulators or pipelined ADC architectures
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**THANK
YOU**

